

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. What are the speculative claims about neural networks that are core to Mechanistic Interpretability (check all that apply)?

1 / 1 point

☒ Features

☒ Correct

Correct! One speculative claim about neural networks is that features are the fundamental unit of neural networks and they correspond to directions.

☒ Circuits

☒ Correct

Correct! One speculative claim about neural networks is that features are connected by weights, forming circuits, which are computational subgraphs of a network.

☒ Universality

☒ Correct

Correct! One speculative claim about neural networks is that analogous features and circuits form across models and tasks (concept of Universality).
2. This is an example of?: "This part of the image contains a dog fur-like texture"

1 / 1 point

☒ Feature

☐ Circuit

☐ Concept

☒ Correct

Correct! "This part of the image contains a dog fur-like texture" is an example of a feature.
3. What concept helps us avoid the curse of dimensionality and allows us to reason about a model without fitting the whole thing in our heads?

1 / 1 point

☐ Linearity

☐ Universality

☒ Decomposability

☒ Correct

Correct! Decomposability, or the concept that network representations can be described in terms of independently understandable features, helps us avoid the curse of dimensionality.
4. Fill in the blank: The _____ states that the neural networks we observe are just simulations of larger networks where every neuron is a disentangled, or independent feature.

1 / 1 point

☒ superposition hypothesis

☐ universality hypothesis

☐ disentanglement hypothesis

☒ Correct

Correct! According to the superposition hypothesis, Neural Networks, as we observe them, are simulations of larger networks, where each neuron is a disentangled feature.
5. Which of the following is NOT a Mechanistic Interpretability method?

1 / 1 point

☐ Perturbation-based methods

☐ Representation learning

☒ Constitutional learning

☒ Correct

Correct! Constitutional learning is not commonly used in Mechanistic Interpretability research.
6. Which of the following are arguments for curve detectors (check all that apply)?

1 / 1 point

☒ Synthetic examples

☒ Correct

Correct! Synthetic examples are used as an argument for curve detectors. Curve detectors respond as expected to synthetic curve images.

☒ Feature Implementation

☒ Correct

Correct! By looking at the circuit constructing the curve detectors, we can read a curve detection algorithm off of the weights.

☒ Joint tuning

☒ Correct

Correct! Joint Tuning can also provide an argument for curve detectors and evidence for features. If we rotate curve images, at some point the neuron stops firing and curve detectors in the next orientation start firing.
7. When a neuron is polysemantic, what does that mean?

1 / 1 point

☒ Polysemantic neurons respond to multiple unrelated inputs

☐ Polysemantic neurons respond to multiple related inputs

☐ Polysemantic neurons respond to only a single input

☒ Correct

Correct! Neural networks often contain polysemantic neurons that respond to multiple unrelated inputs, like a neuron that responds to cat faces, fronts of cars, and cat legs.
8. As we increase sparsity, polysemantic neurons start to emerge, starting with the features with lowest importance. Eventually, all neurons become highly polysemantic. What is this evidence of?

1 / 1 point

☐ A: Universality

☒ Superposition

☐ Circuits

☒ Correct

Correct! This was the result of an experiment from "Toy Models of Superposition" by Elhage, et.al.
9. One approach to Representation Learning is:

1 / 1 point

☒ Dictionary Learning

☐ Tree Ensembles

☐ Prototype-based Learning

☒ Correct

Correct! Dictionary learning is an unsupervised machine learning technique that aims to find a set of basis vectors or atoms that can represent high-dimensional data in a sparse and efficient way. The goal is to learn a dictionary of atoms from the data itself, rather than using a pre-defined set of basis vectors.
10. The first time representation learning was scaled to very large language models was in Spring 2024. What LLM was this done with?

1 / 1 point

☐ Gemini

☐ gpt-4o

☒ Claude 3 Sonnet

☒ Correct

Correct! Anthropic used sparse autoencoders to produce interpretable features for the Claude 3 Sonnet model, which was the first time representation learning was scaled to this level.